RAGHVENDER TYAGI

AI Systems & Distributed Backend Engineer | B.Tech CSE (AI), KIET '27

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EXPERIENCE

Lightning Leap Analytics — AI Research Intern (Remote) | **Jan 2026 – May 2026**

Contributions to Tequity (Enterprise AI Analytics Platform):

- Engineered natural-language query retrieval pipelines and semantic metadata indexing to ground enterprise questions in schema context.
- Developed an agentic multi-agent workflow in LangGraph featuring state persistence, custom domain tools, and dynamic query routing between specialized sub-agents.
- Constructed hybrid Text-to-SQL engine on PostgreSQL (pgvector) with dual-layer AST and regex validator (validate_sql_access) enforcing strict RBAC and preventing raw data exposure.

SELECTED PROJECTS

TakeMyCompute | *Django, Channels, Redis, Docker, PostgreSQL, Celery, WebSockets*

- Distributed P2P compute marketplace allowing hardware providers to monetize idle compute and renters to run untrusted code.
- Enforced containerized isolation with docker-py, applying hard cgroup CPU/RAM limits and an isolated bridge network namespace.
- Implemented real-time system telemetry via psutil over WebSockets and Redis Pub/Sub log streaming with REST fallback.

Paperly AI | *LangGraph, Python, ThreadPoolExecutor, Django, Celery, Redis, arXiv API*

- Autonomous multi-agent research engine orchestrating literature synthesis and academic paper generation.
- Parallelized LLM workers via ThreadPoolExecutor inside LangGraph, slashing end-to-end synthesis latency by 65%.
- Built async Django + Celery + Redis pipeline preventing HTTP timeouts; integrated arXiv API for literature and novelty scoring.

NETRA MITRA | *Raspberry Pi, ESP32-CAM, Ultrasonic Sensors, Decision Tree, Embedded Python*

- AI robotic guidance system for visually impaired people with 8 ultrasonic sensors and decision-tree obstacle avoidance.
- Dual-processor edge pipeline coupling ESP32-CAM video capture with Raspberry Pi scene description and audio cues. Patent filed (2024).

GitHub Change Impact Analyzer | *Python, Static Analysis, AST, Dependency Graphs, ML*

- Pull request intelligence tool analyzing code changes, dependency graphs, and static ASTs to detect downstream blast radius and risk.

TECHNICAL SKILLS

Languages: Java, Python, SQL
Backend: Django, FastAPI, PostgreSQL, Redis, Celery, REST APIs
GenAI & Agents: LangGraph, LangChain, RAG, MCP, Multi-Agent Architectures
Distributed Systems: Docker (cgroups, namespaces), WebSockets (Channels), Redis Pub/Sub, System Monitoring
CS Fundamentals: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

PATENT & CERTIFICATIONS

- **Patent Application Filed (2024):** AI-Powered Robotic Guidance System for Visually Impaired Individuals
- **AWS Certified AI Practitioner** — Amazon Web Services
- **LangGraph** — LangChain Academy

ACHIEVEMENTS & HACKATHONS

- Winner, INNOTECH (KIET Group of Institutions)
- Finalist, OpenAI State Finals (Noida International University)
- Finalist, Techno-Business Finals (Jaipuria Institute of Management)
- Winner, Innovation Project Award (Smart Irrigation System)
- 1st Rank (Department Teams), Smart India Hackathon (Institute-level Selection)

EDUCATION

KIET Group of Institutions, Ghaziabad — B.Tech in CSE (Artificial Intelligence) | **Oct 2023 – 2027**

CGPA: 7.6 / 10.0